

Hoja 3

4

$$DV = 2^{20}$$

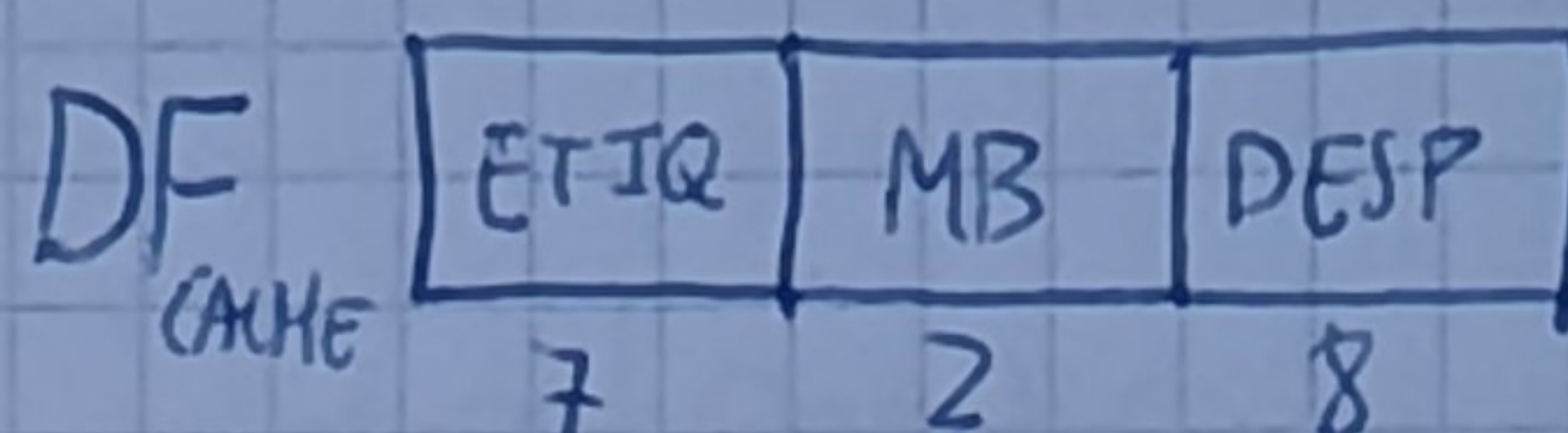
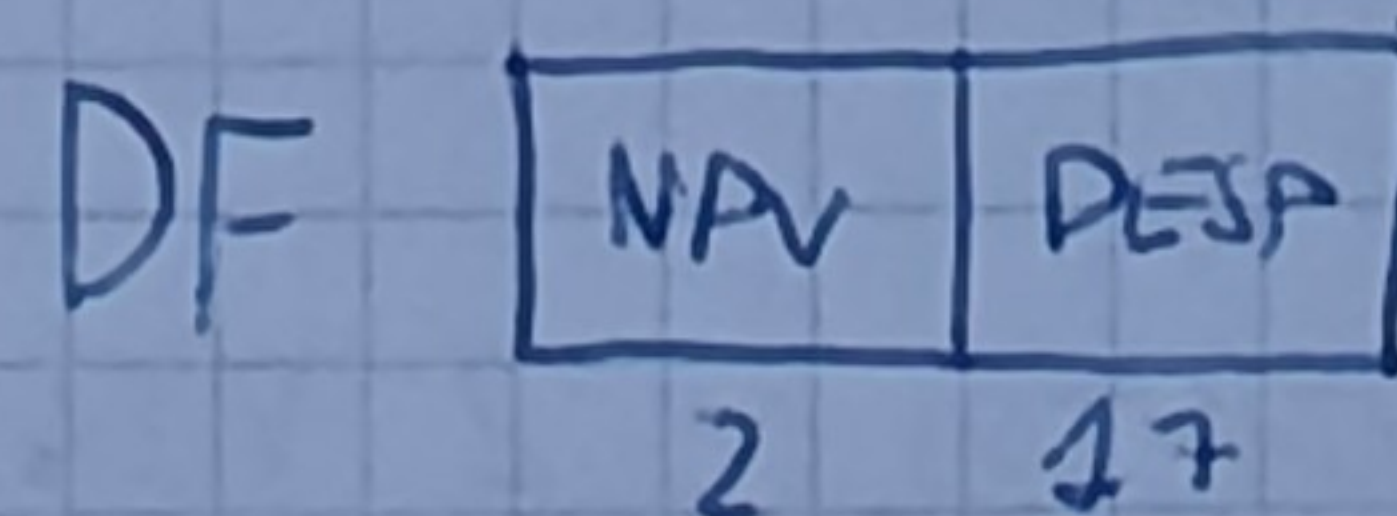
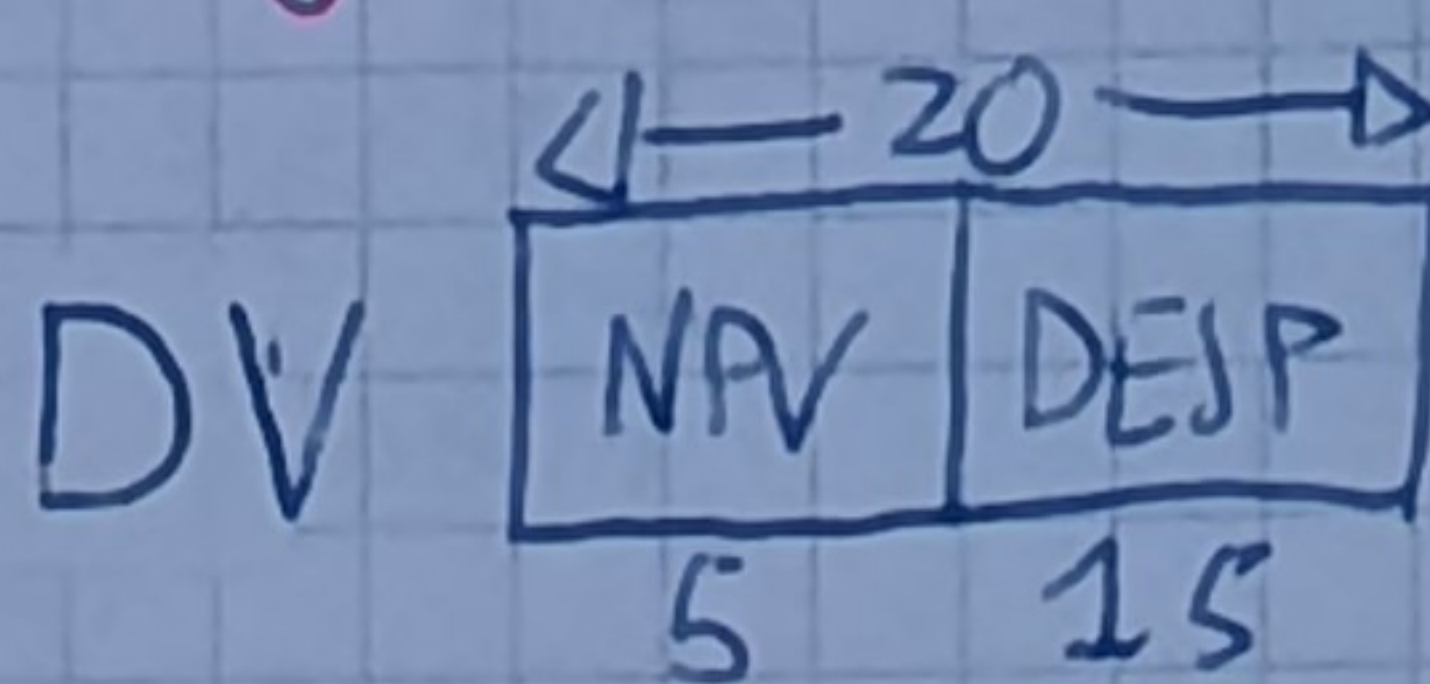
$$MF = 128 \text{ Kb} = 2^{17}$$

$$T_{\text{PAG}} = 32 \text{ Kb}, LRU = 2^{15}$$

$$MC = 1 \text{ Kb}$$

EMP Directo

$$T_{\text{bloque}} = 256 \text{ b}$$



a, b)

CACIÓN	DIRECCIÓN	DIR. BINARIA	NPV
1	0x50000	0101 0000 0000 0000 0000	0x0A
2	0xF4000	1111 0100 0000 0000 0000	0x1E
3	0x66000	0110 0110 0000 0000 0000	0x0C
4	0x12000	0001 0010 0000 0000 0000	0x02
5	0x40000	0100 0000 0000 0000 0000	0x08
6	0x36000	0011 0110 0000 0000 0000	0x06

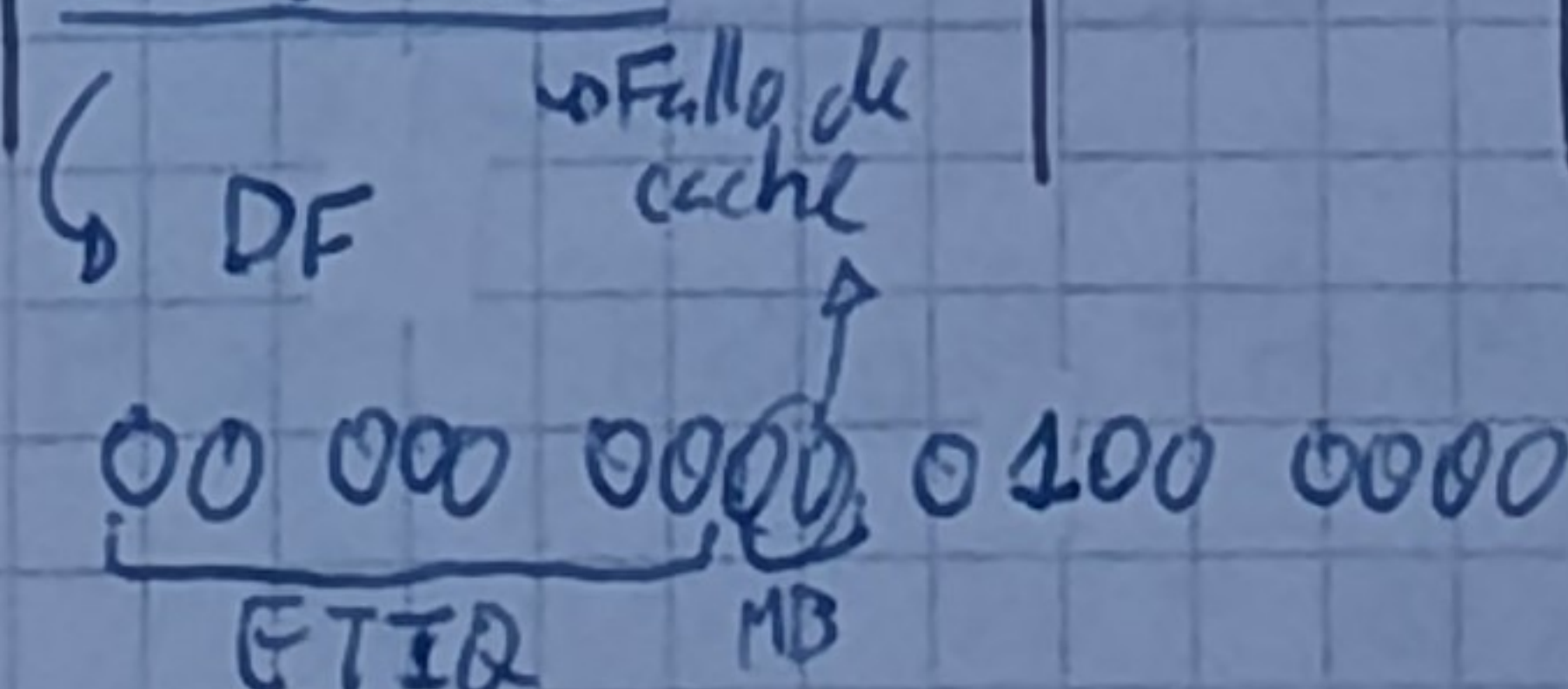
$$8 \text{ Kb} = 2^{13} \rightarrow 1111 \ 1111 \ 1111 \ 1111 \ 1111 \ \checkmark \text{ (no se sale)}$$

c)

TLB		CACHE		ETIQ / BL BINARIO	dir física		dir virtual	ACCIÓN
NPV	NPF	ETIQ	BLOQUE		PPE	NPV		
0x0A	1	0x70	0	111 0000 00	11	0x1E 111 10	2	
0x06	2	0x27	1	010 0111 01	01	0x0A 01010	1	
0x02	0	0x27	2	010 0111 10	01	01010	1	
0x1E	3	0x27	3	010 0111 11	01	01010	1	

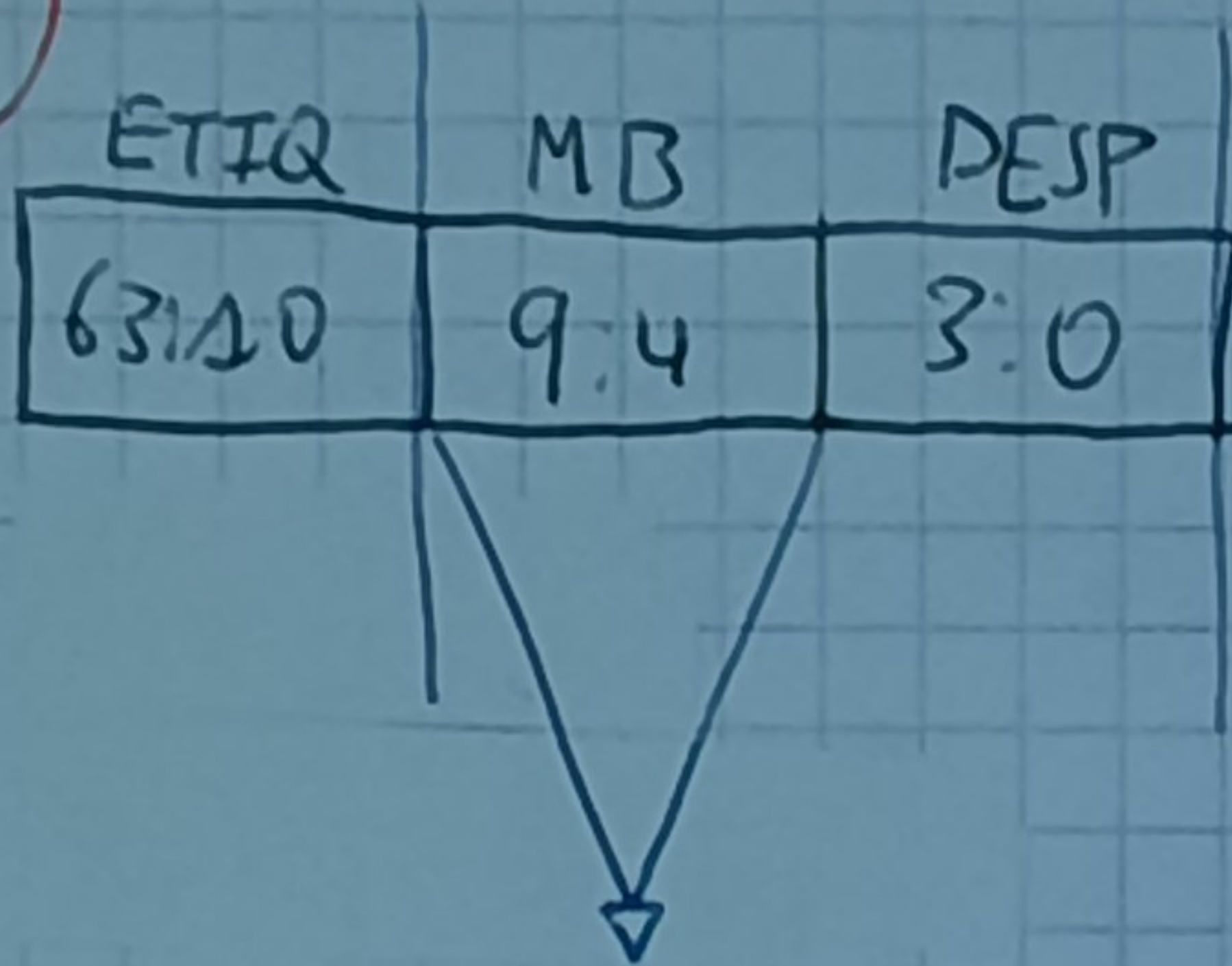
d)

REF	DIRECCIÓN	DIR. BINARIA	NPV	NPF	DIR. ES
1	0xF40FS	1111 0100	0x1E	3	11 10000000 - ALERTA
2	0x66000	0110 0110 0000	0x0C		
3	0x40040	0100 0000 0000			
4		0100 0000			



CACHE	BL
0x00	
0x70	0

1



$$2^{10} = 1 \text{ KB}$$

$$T_{\text{bloque}} = 16 \text{ bytes}$$

$$T_{\text{PAG}} = 2^{10}$$

$$CJTD \Rightarrow 2^6 \text{ CONJUNTOS} \times 2 \text{ VIAS} = 2^7 \text{ BLOQUES} \times 2^4 \frac{\text{bytes}}{\text{bloque}} = 2^{11} \text{ bytes}$$

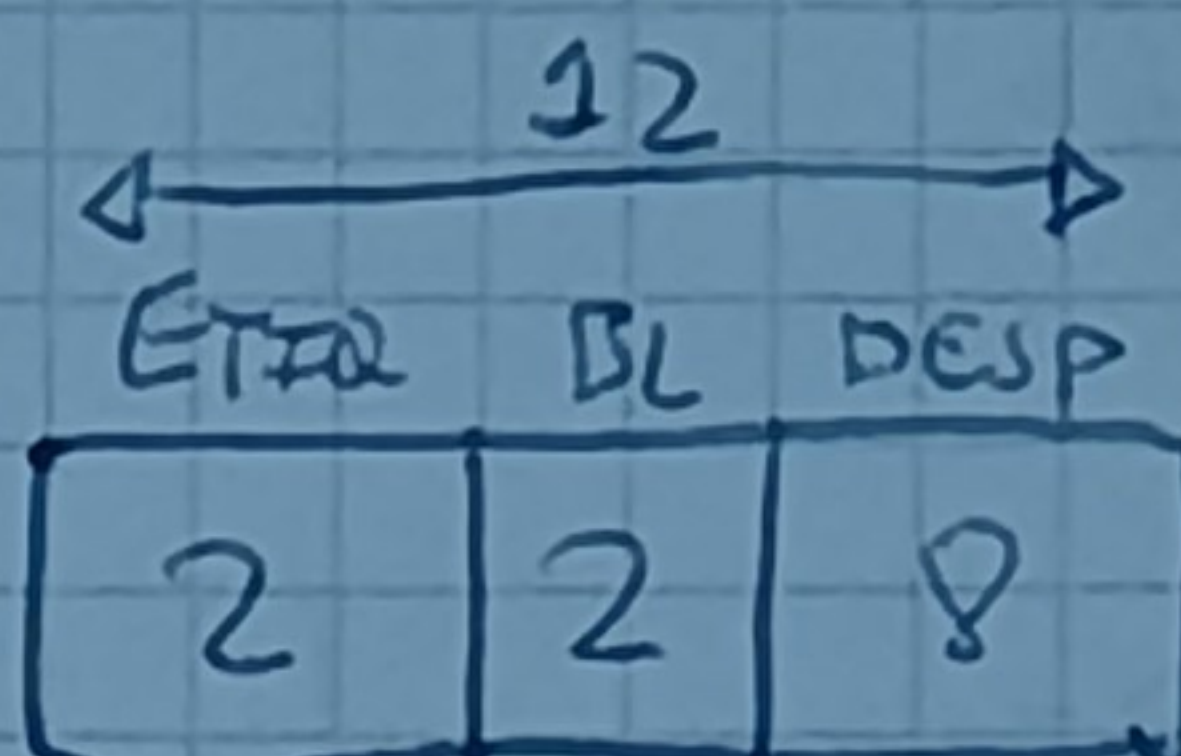
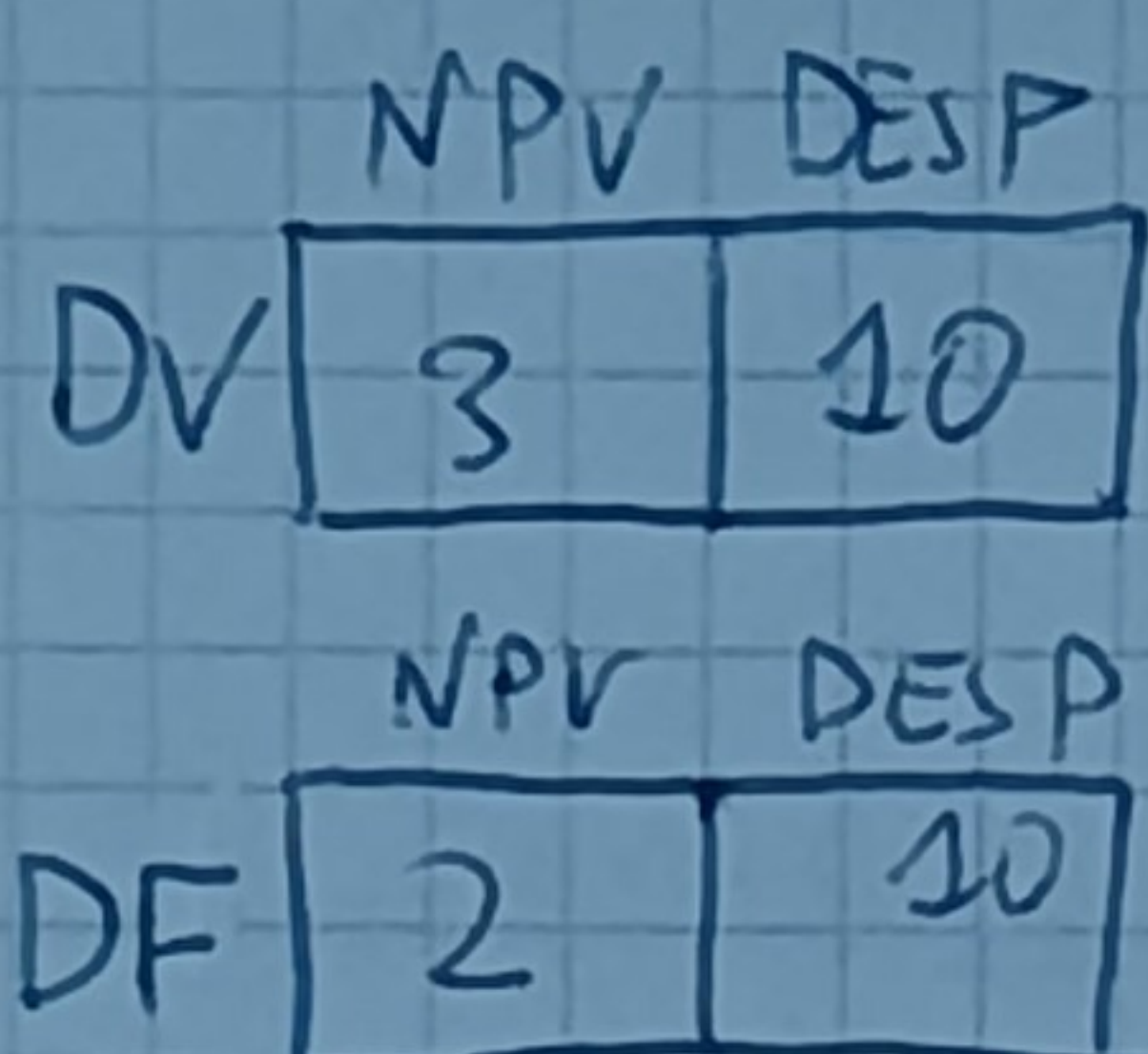
2

$$T_{\text{PAG}} = 1024 \text{ pág}$$

$$MV \rightarrow 8 \text{ PAG}$$

$$MF \rightarrow 4 \text{ PAG}$$

$$N^{\circ} \text{ bloques} \rightarrow 4$$



PAG VIR	VAL	PAG MARC
0	1	3
1	1	1
2	0	0
3	0	0
4	1	2
5	0	0
6	1	0
7	0	0

$$T_{\text{PAG}} = 1024 \text{ pág}$$

$$N^{\circ} \text{ Bloq} = 4$$

$$\left. \begin{array}{l} T_{\text{PAG}} = 1024 \text{ pág} \\ N^{\circ} \text{ Bloq} = 4 \end{array} \right\} \frac{1024}{4} = 256 \text{ pág/bloque}$$

$$PV=0 \Rightarrow PF=3 \Rightarrow \text{DIR FÍSICA} \Rightarrow \text{11xx xxxxx xxxxx}$$

3

$$MV = 16 \text{ Mb} = 2^{24} \text{ bytes}$$

$$T_{\text{PAG}} = 4 \text{ KB} = 2^{12} \text{ bytes}$$

$$MP = 32 \text{ KB} = 2^{15} \text{ bytes}$$

$$MC = 8 \text{ KB} = 2^{13} \text{ bytes}$$

$$T_{\text{bl}} = 64 \text{ bytes} = 2^6 \text{ bytes}$$

$$ASOC \times CJTD = 2$$

$$\text{int } A[16384]$$

$$\text{for } (i=0; i < 16384; i++) \{$$

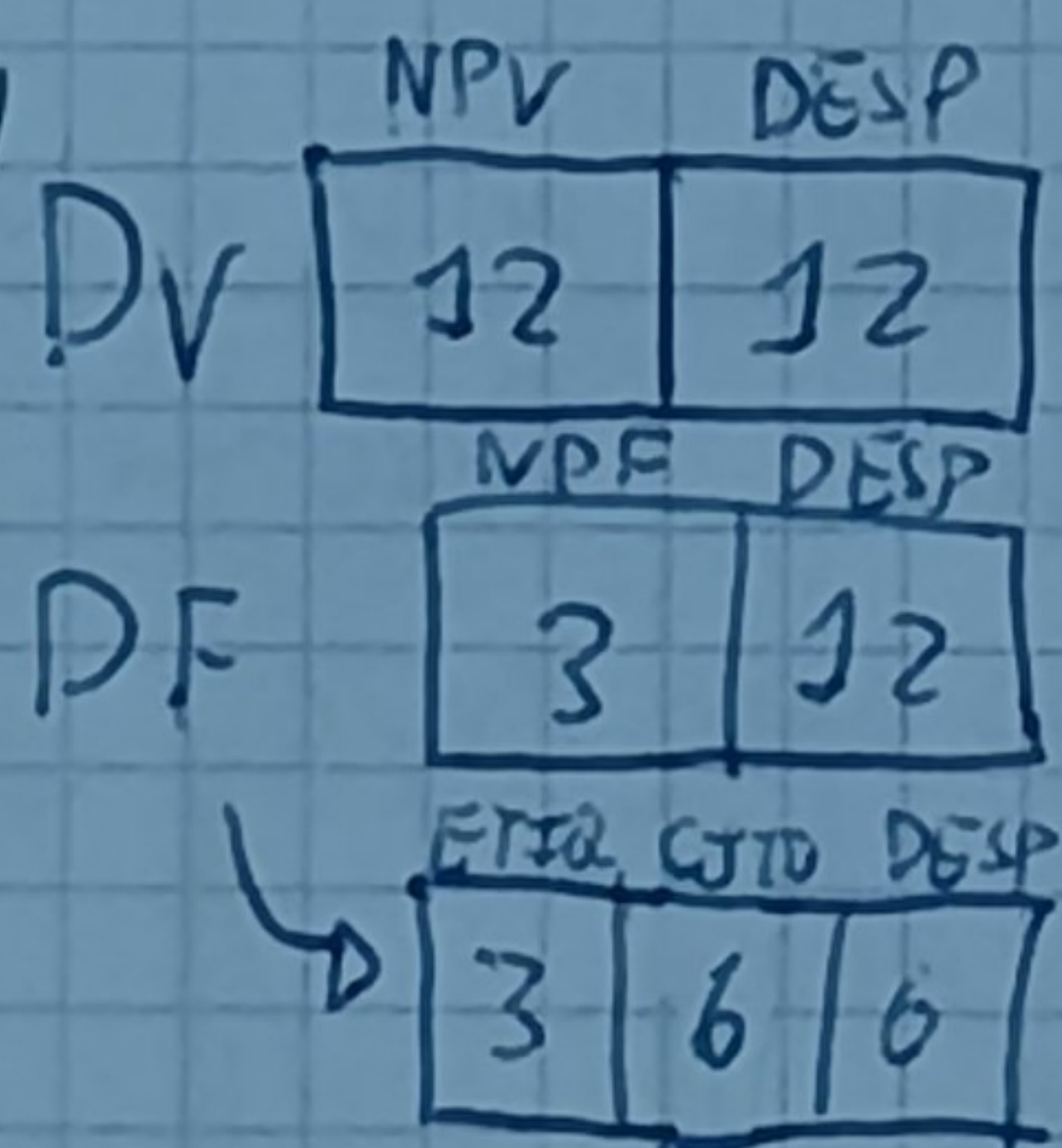
$$\quad \text{val} += A[i];$$

$$\}$$

$$DV = 0x100000$$

2⁷ bloques } 6 CJTD

NPV	NPF	LRV
100	0	0
101	1	1
200	2	2
201	3	3
300	4	4
301	5	5
400	6	6
401	7	7



$$TAM A = 16384 \times 4 \text{ bytes} = 2^{16} \text{ bytes}$$

$$\frac{2^{16} \text{ bytes}}{2^{12} \text{ bytes/pág}} = 16 \text{ pág} \Rightarrow 0x100 + 0x10F$$

2cientos, 14 follos

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$$MV = 32 \text{ pgs} \times 8 \text{ Kb} = 256 \text{ Kb} = 2^8 \cdot 2^{10} = 2^{18} \text{ bytes}$$

$$MF = 32 \text{ Kb} = 2^5 \cdot 2^{10} = 2^{15} \text{ bytes}$$

$$MC = 512 \text{ bytes} = 2^9 \text{ bytes}$$

ASOC 2 VIAS

$$T_{\text{bloque}} = 128 \text{ bytes} = 2^7 \text{ bytes}$$

c) DV

NPV	DESP
5	13

DF

NPV	DESP
2	13

ETIQ	CTO	DESP
7	1	7

b) T.P. INV

NPV	NPF	NPF	Dir Virtual	P.V. HEX	Dir física HEX
06	3	3	0011 0xx...x	0C000-0DFFF	14xx...x 6000-7FFF
01	0	0	0000 1xxx...	02000-03FFF	00xxx... 0000-1FFF
04	2	2	0010 0xxx...	08000-09FFF	10xxx... 4000-5FFF
0C	1	1	01100xxx...	18000-19FFF	01xxx... 2000-3FFF

CACHE

ETIQ	CTO	VIA	DF	ETIQ	CTO	NPV
23	0	0	①	010 0011	0xxx xxxxx	2300-237F 01100 18300-1837F
47	0	1	②	100 0111	0xxx xxxxx	4700-477F 00100 08700-0877F
73	1	0	③	111 0011	1xxx xxxxx	7380-73FF 00110 0D380-0D3FF
08	1	1	④	000 1000	1xxx xxxxx	0880-08FF 00001 02880-028FF

NPV	NPF	REF	NPV	NPF	REF	NPV	NPF
06	3	0x08770 NPV=00100 NPF=10	0x02080 NPV=00001 NPF=00	0x02880 NPV=00001 NPF=00	0x0D3F0 NPV=00110 NPF=11	0x27000 NPV=10011 NPF=01	
01	0		01(A) 0	01(A) 0			
04	2	04(A) 2					
0C	1					13(F) 1	
ETIQ	CTO/VIA						
23	0/0				74(F) 0/0	74 0/1	
47	0/1	47(A) 011			47 0/1	30(F) 0/1	
73	1/0		00(F) 1/0		73(I) 1/0		
08	1/1			08(A) 1/1			

	NPV	NPF	DIR FÍSICA	FALLOS	ACIERTOS
0x08770	00100	10	100 0111 0111 0000	1	31
0x02080 - 02009F	00001	00	000 0000 1000 0000	0	32
0x02880 - 0289F	00001	00	000 1000 1000 0000	1	15
0x0D3F0 - 0x0D410	00110	11	111 0011 1111 0000	1	16
0x27000 - 0x2701F	10011	01	011 0000 0000 0000	1	31

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$$MV = 4GB = 2^{32}$$

$$TPAG = 2Kb = 2^{11}$$

$$MP = 2^{13}$$

$$MC = 2048 \text{ bytes} = 2^{11}$$

$$T_{\text{bloque}} = 256 \text{ bytes} = 2^8$$

ASOC 4 VÍAS

a)

	NPV	DESP	
DV	21	11	
	NPF	DESP	
DF	2	11	
	ETQR	CJTO	DESP
	4	1	8

b)

NPV	MARCO PAG	EDAD
0x022C02	0	0x2
0x1B0000	1	1x3
0x0AA231	2	3x1
0x100000	3	2x0

$$0x5518FF8 \rightarrow 0101 \ 0101 \ 0001 \ 0001 \ 1000 \ 1111 \ 1111 \ 10$$

$$NPV = 0x0AA231$$

$$DF = 1 \ 0111 \ 1111 \ 1000 \ \text{ACTEIZTO}$$

$$0xA000000 \rightarrow 0000 \ 1010 \ 0000 \ 0000 \ 0000 \ 0000 \ 0000 \ 0000$$

NPV = 0x014000 → Fallo de página, se trae página a mano de pág 3

$$DF = 1 \ 1000 \ 0000 \ 0000$$

$$0x1281276 \rightarrow 0001 \ 0001 \ 0010 \ 1000 \ 0001 \ 0101 \ 0111 \ 0110$$

NPV = 0x022502 → fallo de página

$$NPF = 01$$

No afecta a la cache → No hay que invalidar bloque

Se elimina NPV 0x1B0000 → $DF = 0 \ 1xxx \ xxxx \ xxxx \rightarrow 0 \ 1010 \ 0111 \ 0110$

CJTO	BL	ETQR	EDAD	VAL
0	0	C	1	1
	1	0	3	1
	2	5	0	1
	3	1	2	1
1	0	A	1	
	1	3	3	
	2	1	2	
	3	B	0	

c) 0xFFA34500

$$1111 \ 1111 \ 1010 \ 0011 \ 0100 \ 0101 \ 0000 \ 0000$$

NPV = 0x1FF468 → fallo de página

$$DV = \begin{cases} 0xFFA34000 \\ 0xFFA347FF \end{cases}$$

$$NPF = 01$$

$$DF = 0 \ 1xxx \ xxxx \ xxxx$$

$$0x0100 - 0x0FFF$$

$$DF = 0 \ 1101 \ 0000 \ 0000$$

$$DF = 0 \ 1101 \ xxxx \ xxxx$$

$$0x0D00 - 0x0DFF$$

$$DV = \begin{cases} 0xFFA34500 \\ 0xFFA345FF \end{cases}$$

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for (j=0; j<512; j++)
  for (i=0; i<512; i++)
    red[j] = red[j] + A[i][j]
```

$A[i][j] = 0 \times 0045800$

Tamaño $A = 2^9 \cdot 2^9 \cdot 2^2 = 2^{20}$ bytes

$N^\circ \text{ p\u00e1g} = \frac{2^{20}}{2^{12}} = 2^8 = 256 \text{ p\u00e1g}$

b) 1\u00b0 iteraci\u00f3n:

$\text{red}[0](F), A[0][0](F)$

$\text{red}[0](A), A[1][0](A)$

$\text{red}[0](A), A[2][0](F)$

$N^\circ \text{ Ref } A = 512 \times 512 = 262144$

$N^\circ \text{ folios } A = \frac{262144}{2} = 131072$

$N^\circ \text{ total de folios} = 131073$

a)

32	
NPV	DESP
20	12
NPF	DESP
xxx	12

$NPV_{\text{m\u00edd}} A = 0 \times 00458$

Cada p\u00e1g 2 folios de la matriz

$\begin{cases} A[0][0-511] \\ A[1][0-511] \end{cases} \Rightarrow \text{\u00faltima p\u00e1g } A[2][0-511] = 0 \times 005$
 $\text{red} \Rightarrow 0 \times 00558 \text{ ocupa la m\u00edd}$

Acierto $A = 131072$

Acierto, $\text{red} = 262143 \text{ lecturas}$
 $= 262144 \text{ escrituras}$

(=) intercambio bucles

$\hookrightarrow \text{Total folios} = 256 + 1$

$A = 0 \times 00128000$

$B = 0 \times 00941600 - 0 \times 00941400$

$C = 0 \times 00002000$

$D = 0 \times 005110A0 - 0 \times 005114A0$

$E = 0 \times 00300400 - 0 \times 00300500$

$F = 0 \times 00041000$

8

DV	NPV	DESP
	20	12
DF	NPF	DESP
	16	12

C\u00f3digo 3500B $\rightarrow 0 \times 000C0000$

M\u00e1r	P\u00e1g	C\u00d3DIGO	A	B	A	C	D	A	E	//
6		$0 \times 000C0(F)$	$0 \times 000C06$	$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$	
7			$0 \times 00128(F)(1)$	$0 \times 00128(1)$	$0 \times 00128(0)(A)$	$0 \times 00128(1)$	$0 \times 00128(2)$	$0 \times 00128(0)(A)$	0×00128	//
8				$0 \times 00941(6)(F)$	$0 \times 00941(1)$	$0 \times 00941(2)$	$0 \times 00511(1)(6)$	$0 \times 00511(1)$	0×00511	
9							$0 \times 00002(6)(F)$	$0 \times 00002(1)$	$0 \times 00002(2)$	$0 \times 00300(F)$
TLB										//
0		$0 \times 000C0$	$0 \times 000C0 \rightarrow 6$	$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$	
1			$0 \times 00128 \rightarrow 7$	$0 \times 00941(F)$	$0 \times 00128(F)$	$0 \times 00002 \rightarrow 9$	$0 \times 00511 \rightarrow 8$	$0 \times 00128 \rightarrow 7$	$0 \times 00300 \rightarrow 9$	//
			B	A						
6			$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$					
7			0×00128	$0 \times 00128(0)(A)$	$0 \times 00128(0)(A)$					
8			$0 \times 00941(F)$	$0 \times 00941(2)$	$\Rightarrow 0 \times 00041(1)$					
9			0×00300	$0 \times 00300(1)$	$0 \times 00300(2)$					
0			$0 \times 000C0$	$0 \times 000C0$	$0 \times 000C0$					
1			$0 \times 00941 \rightarrow 8$	$0 \times 00128 \rightarrow 7$	$0 \times 00041 \rightarrow 8$					